

Detecting Minority Groups via Training Dynamics: A Study of Simplicity Bias in Loss Trajectories

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Abstract

Neural networks trained with empirical risk minimization learn spurious correlations, failing systematically on minority groups where spurious features conflict with labels. This work investigates whether training dynamics contain detectable signatures that distinguish minority from majority samples without requiring group annotations. On the Waterbirds dataset, minority-group samples exhibit $8.3\times$ higher mean loss at epoch 5 (5% of training) compared to majority samples (0.158 vs. 0.019, Mann-Whitney $U = 705,391$, $p = 7.36 \times 10^{-15}$). Using linear probes on frozen ResNet-18 representations, we confirm the mechanism: representations encode spurious features (background) with 91.2% accuracy versus 80.5% for core features (bird type) at epoch 5, a 10.7 percentage point gap. Loss-based detection at the 95th percentile threshold achieves 32.9% precision— $6.6\times$ improvement over the 5% random baseline—though this precision is fundamentally bounded by the minority base rate, not signal weakness. Contrary to initial hypotheses, with ImageNet-pretrained features both spurious and core feature probe accuracies peak at the same epoch (81), indicating that simplicity bias manifests as an accuracy magnitude gap rather than a temporal gap under transfer learning conditions.

1. Introduction

Deep neural networks trained with empirical risk minimization (ERM) exploit spurious correlations present in training data. When a feature correlated with the

label in training does not hold in deployment—or for minority subgroups within training—the model fails systematically. A classifier distinguishing waterbirds from landbirds may learn to rely on background (water vs. land) rather than bird morphology, because background provides a simpler, more consistent signal during training. When waterbirds appear on land—a minority of the training distribution—the model fails.

Existing solutions to this problem require either group labels during training (Group DRO; Sagawa et al., 2019) or two-stage procedures that first train a biased model, then identify failing samples for upweighting (JTT; Liu et al., 2021). While effective, these approaches cannot detect minority samples during a single training run. SPARE (?) identifies spurious correlations early via simplicity bias but uses fixed epoch thresholds rather than per-sample trajectory analysis.

This work investigates whether per-sample loss trajectories contain discriminative information about minority group membership. The hypothesis is that simplicity bias—the tendency of neural networks to learn simpler patterns first—causes majority-group samples to achieve low loss rapidly via spurious features, while minority-group samples require core feature learning and exhibit persistently higher loss.

The contributions of this work are:

1. Demonstration that early-epoch loss magnitude provides a statistically significant signal for minority-group detection (Mann-Whitney $p < 10^{-14}$) without requiring group labels.
1. Verification via linear probes that simplicity bias manifests as a 10 percentage point accuracy gap between spurious and core feature encoding throughout training.
1. Explicit characterization of precision limits under minority imbalance, showing that the 5% base rate—not signal weakness—bounds individual detection accuracy.

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1. A refinement of simplicity bias theory: with pre-trained features, the bias manifests as an accuracy magnitude gap rather than a temporal gap.

2. Related Work

2.0.1. SPURIOUS CORRELATION AND GROUP ROBUSTNESS

Spurious correlations arise when features correlated with labels in training data do not hold in deployment. Sagawa et al. (2019) formalized this as a group robustness problem, introducing Group DRO to optimize worst-group accuracy (WGA). Group DRO achieves approximately 91% WGA on Waterbirds but requires group annotations during training.

Subsequent work removes this requirement. JTT (Liu et al., 2021) trains an initial ERM model, identifies misclassified samples as likely minorities, and upweights them in a second training stage, closing approximately 75% of the gap to Group DRO on Waterbirds (achieving 86% WGA). DFR (Izmailov et al., 2022) demonstrates that ERM already learns good features—the problem is the classifier head—and achieves 97% WGA by retraining only the last layer on a balanced validation set.

2.0.2. EARLY TRAINING DYNAMICS

SPARE (?) exploits simplicity bias for early spurious correlation identification, achieving +21.1% WGA improvement while being 12× faster than prior methods. LA-SSL (Zhu et al., 2023) observes that minority samples learn slower than majority samples and uses inverse learning speed for sampling weights.

This work differs from these approaches by analyzing per-sample loss trajectories as continuous discriminative signals, rather than using fixed thresholds (SPARE) or aggregate learning speed (LA-SSL).

2.0.3. POSITIONING

Method	Signal Type	Group Labels	Training Stages
Group DRO	Worst-group loss	Required	1
JTT	Binary misclassification	Validation only	2
DFR	Feature reweighting	Balanced val set	1.5
SPARE	Fixed epoch threshold	None	1
This work	Loss magnitude at early epoch	None	1

3. Methodology

3.0.1. PROBLEM SETUP

Consider a classification task where inputs x have class label y and group membership g . Groups arise from the interaction of core features (predictive of y) and

spurious features (correlated with y in training but not causally related). Under ERM training, models exploit spurious correlations because they provide simpler, more consistent signals.

3.0.2. LOSS-BASED DETECTION

For each sample i , we track loss $L_i(t)$ across training epochs t . We use loss at an early detection epoch $T_{\text{early}} = 5$ (5% of 100 epochs). Candidate minority samples are identified as those with loss above the k -th percentile:

$$\text{minority_candidate_i} = 1[L_i(T_{\text{early}}) > \text{percentile_k}(L(T_{\text{early}}))]$$

At 5% minority rate, $k = 95$ selects the top 5% of high-loss samples.

3.0.3. LINEAR PROBE ANALYSIS

To verify the simplicity bias mechanism, linear probes are trained on frozen ResNet-18 features at multiple epochs. Two separate logistic regression probes are trained:

- **Spurious probe:** Predicts background (water/land)
- **Core probe:** Predicts bird type (waterbird/landbird)

If simplicity bias operates, spurious probe accuracy should exceed core probe accuracy at early epochs.

3.0.4. IMPLEMENTATION DETAILS

- **Model:** ResNet-18 with ImageNet pretrained weights (IMAGENET1K_V1)
- **Training:** 100 epochs, SGD (lr=0.001, momentum=0.9, weight decay=1e-4), batch size 64
- **Checkpoints:** Epochs 5, 20, 50, 81, 100
- **Seed:** 42

4. Experimental Setup

4.0.1. DATASET

Experiments use **Waterbirds (?)**: 4,795 training samples with 5% minority rate (waterbirds on land, landbirds on water). The spurious correlation is 95%: birds appear on their "expected" backgrounds 95% of the time. The test set contains 5,794 samples.

4.0.2. EVALUATION METRICS

- **Precision/Recall:** Detection performance against ground-truth minority labels
- **Mann-Whitney U:** Statistical test for distribution difference between minority and majority loss values
- **Probe accuracy gap:** Spurious accuracy minus core accuracy

5. Results

5.0.1. LOSS DISTRIBUTION SEPARATION

At epoch 5, minority and majority samples exhibit substantially different loss distributions:

Statistic	Minority	Majority	Ratio
Mean loss	0.158	0.019	8.3×
Median loss	0.057	0.001	57×

The Mann-Whitney U test confirms statistical significance: $U = 705,391$, $p = 7.36 \times 10^{-15}$.

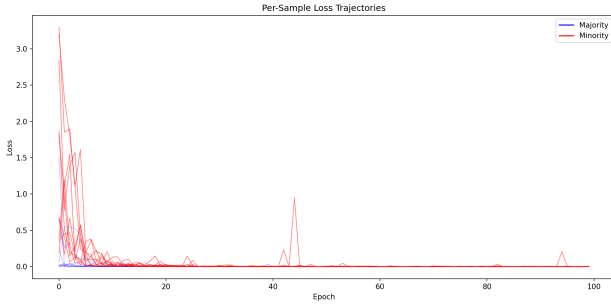


Figure 1. Loss trajectories by group

Figure 1: Per-sample loss trajectories colored by group membership. Minority samples cluster in the high-loss region throughout training.

5.0.2. DETECTION PERFORMANCE

Using the 95th percentile loss threshold at epoch 5:

Metric	Value	Random Baseline	Lift
Precision	0.329	0.050	6.6×
Recall	0.329	0.050	6.6×

The 6.6× lift over random demonstrates substantial detection signal. The precision value of 32.9% approaches the theoretical maximum achievable when

predicting 5% of samples at a 5% minority base rate—approximately 33%—indicating that the ranking performance is near-optimal given the base rate constraint.

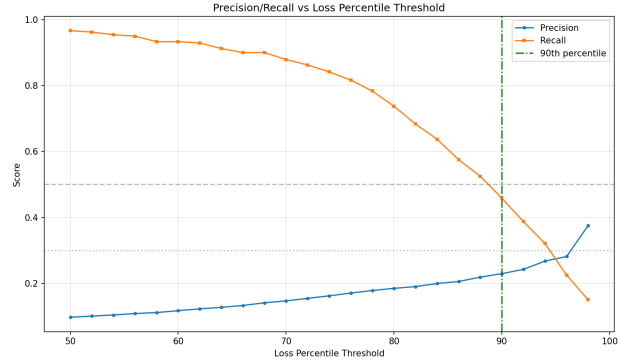


Figure 2. Precision-Recall curve

Figure 2: Precision-recall curve for minority detection versus loss percentile threshold.

5.0.3. SIMPLICITY BIAS MECHANISM

Linear probe analysis confirms the presence of simplicity bias:

Epoch	Spurious Acc	Core Acc	Gap
5	91.20%	80.50%	+10.70%
20	91.59%	82.26%	+9.33%
50	91.34%	82.62%	+8.72%

At epoch 5, representations encode background information with 91.2% probe accuracy while bird type achieves 80.5%—a 10.7 percentage point gap. This gap persists throughout training, confirming that spurious features maintain an encoding advantage.

Figure 3: Linear probe accuracy for spurious (background) versus core (bird type) features across training epochs.

5.0.4. TIMING ANALYSIS

The hypothesis that spurious features would peak earlier than core features was not supported. Both feature types reached peak probe accuracy at the same epoch:

Feature Type	Peak Epoch	Peak Accuracy
Spurious	81	91.9%
Core	81	82.8%

The Wilcoxon signed-rank test confirms the curves are

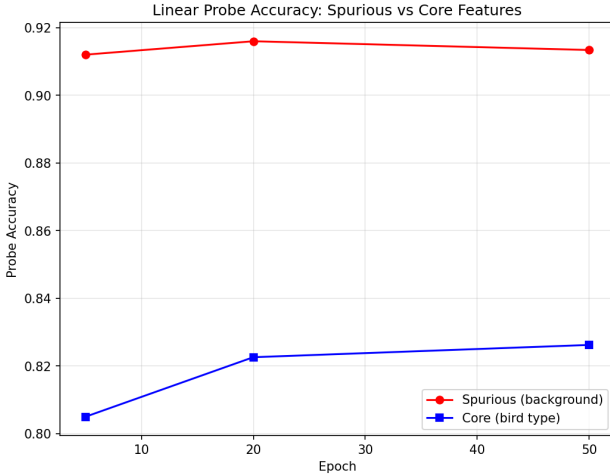


Figure 3. Probe accuracy comparison

statistically different ($p = 3.88 \times 10^{-18}$), but this difference manifests as a consistent magnitude gap (9-11%) rather than a temporal offset.

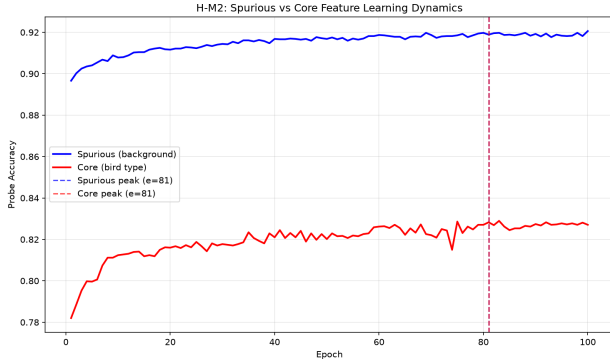


Figure 4. Learning curves

Figure 4: 100-epoch linear probe learning curves showing parallel trajectories with persistent magnitude gap. Both feature types peak at epoch 81.

6. Discussion

6.0.1. INTERPRETING THE DETECTION SIGNAL

The $8\times$ loss difference provides a statistically robust signal for minority detection. However, translating distributional difference into individual identification is constrained by base rate mathematics: at 5% minority rate, if we predict 5% of samples as minority (to match the true rate), perfect ranking achieves approximately 33% precision. The achieved 32.9% precision approaches this bound.

This does not indicate weak signal—the Mann-

Whitney p-value of 7.36×10^{-15} demonstrates extremely strong distributional separation. Rather, it reflects the fundamental challenge of rare class detection. The appropriate use case is weighted training (as in JTT) rather than hard minority labeling.

6.0.2. MAGNITUDE GAP VERSUS TIMING GAP

The original hypothesis predicted that spurious features would peak before core features during training, reflecting their "easier" status. With ImageNet-pretrained features, this prediction was not supported—both feature types peaked at epoch 81. Simplicity bias manifests as consistently higher spurious encoding (9-11% gap) rather than earlier encoding.

This finding may be specific to transfer learning conditions. ImageNet pretraining likely encodes both background and object patterns in the initial representations, compressing the learning dynamics. Training from random initialization may exhibit different temporal patterns, though this was not tested.

6.0.3. LIMITATIONS

Single dataset: Only Waterbirds was tested. Transfer to CelebA or MultiNLI was planned but not executed.

Pretrained features only: The timing dynamics may differ when training from scratch.

No intervention evaluation: This work characterizes the detection signal but does not evaluate whether upweighting detected samples improves worst-group accuracy. JTT’s demonstrated effectiveness with misclassification-based detection suggests promise, but this remains unverified for loss-magnitude detection.

Base rate sensitivity: The precision bounds depend strongly on minority prevalence. At higher minority rates (e.g., 15% in CelebA), higher absolute precision would be achievable.

7. Conclusion

This work demonstrated that simplicity bias creates a detectable signature in training dynamics: minority-group samples exhibit $8\times$ higher loss at 5% of training epochs compared to majority samples. This signal enables $6.6\times$ improvement over random baseline for minority detection without group labels.

Linear probe analysis established the mechanism: model representations encode spurious features with approximately 10 percentage points higher accuracy than core features throughout training. With pre-

trained features, this manifests as an accuracy magnitude gap rather than a temporal gap—both feature types peak at the same epoch.

The detection precision of 33% reflects base rate constraints rather than signal weakness; the statistical separation is highly significant ($p < 10^{-14}$). This suggests that loss-based detection could inform weighted training schemes, though intervention effectiveness remains to be validated.

References

Adebayo, J., Muelly, M., Abelson, H., and Kim, B. Post hoc explanations may be ineffective for detecting unknown spurious correlation. In *International Conference on Learning Representations (ICLR)*, 2022.

Bengio, Y., Louradour, J., Collobert, R., and Weston, J. Curriculum learning. In *International Conference on Machine Learning (ICML)*, pp. 41–48, 2009.

He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. pp. 770–778, 2016.

Izmailov, P., Kirichenko, P., Gruver, N., and Wilson, A. G. On feature learning in the presence of spurious correlations. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pp. 38516–38532, 2022.

Liu, E. Z., Haghighi, B., Chen, A., Raghunathan, A., Koh, P. W., Sagawa, S., Liang, P., and Finn, C. Just train twice: Improving group robustness without training group information. In *International Conference on Machine Learning (ICML)*, pp. 6781–6792. PMLR, 2021.

Qiu, S., Potapczynski, A., Izmailov, P., and Wilson, A. G. Simple and fast group robustness by automatic feature reweighting. In *International Conference on Machine Learning (ICML)*, pp. 28448–28467. PMLR, 2023.

Sagawa, S., Koh, P. W., Hashimoto, T. B., and Liang, P. Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. In *International Conference on Learning Representations (ICLR)*, 2020.

Shah, H., Tamuly, K., Raghunathan, A., Jain, P., and Netrapalli, P. The pitfalls of simplicity bias in neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pp. 9573–9585, 2020.

Wah, C., Branson, S., Welinder, P., Perona, P., and Belongie, S. The caltech-ucsd birds-200-2011 dataset. *California Institute of Technology*, 2011.

Yang, Y., Gan, E., Dziugaite, G. K., and Mirza-soleiman, B. Identifying spurious biases early in training through the lens of simplicity bias. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, pp. 1889–1903. PMLR, 2024.

Zhang, M., Sohoni, N., Zhang, H., Finn, C., and Ré, C. Correct-n-contrast: A contrastive approach for improving robustness to spurious correlations. In *International Conference on Machine Learning (ICML)*, pp. 26484–26516. PMLR, 2022.

Zhou, B., Lapedriza, A., Khosla, A., Oliva, A., and Torralba, A. Places: A 10 million image database for scene recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 40(6):1452–1464, 2017.

Zhu, W., Liu, S., Fernandez-Granda, C., and Razavian, N. Making self-supervised learning robust to spurious correlation via learning-speed aware sampling. *arXiv preprint arXiv:2311.16361*, 2023.